

- 4 *Trypanosoma brucei* is a unicellular organism that causes the infectious disease known as sleeping sickness. Insects known as tsetse flies pass on the organism from infected people to uninfected people when male and female tsetse flies feed on human blood.

(a) Fig. 4.1 is a transmission electron micrograph of the form of *T. brucei* found in human blood.

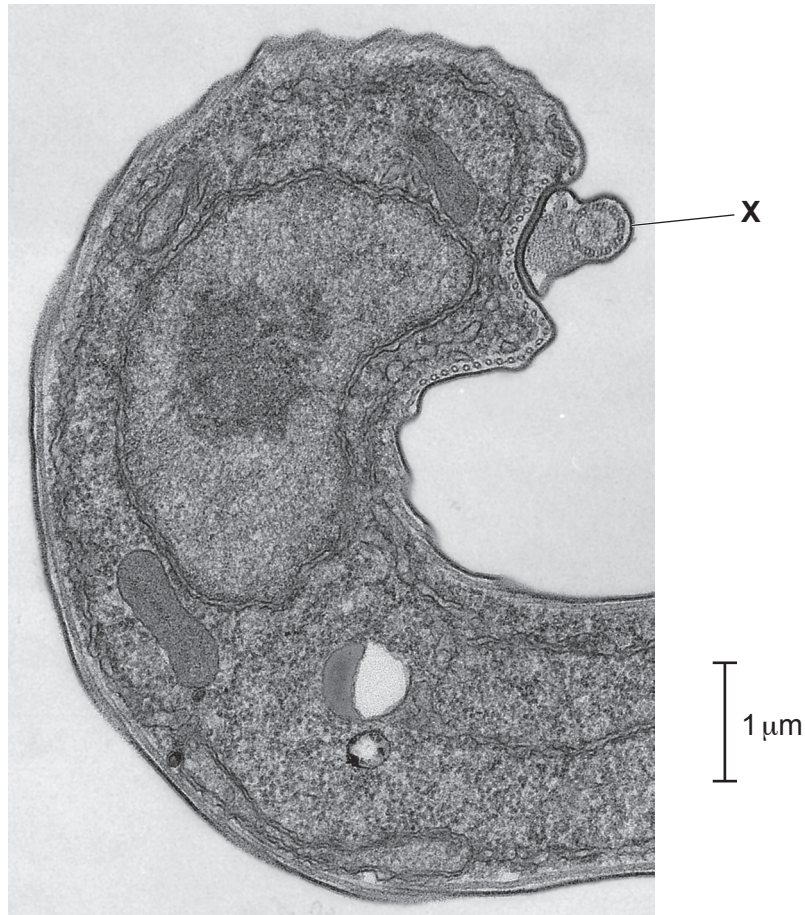


Fig. 4.1

- (i) Draw an arrow on Fig. 4.1 to indicate the location in the cell where ribosomal RNA (rRNA) and proteins are assembled to make ribosomal subunits. [1]
- (ii) With reference to Fig. 4.1, explain how the structure labelled X provides evidence that *T. brucei* is motile (able to move).

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..... [2]

(iii) With reference to Fig. 4.1, explain whether *T. brucei* is a eukaryote or prokaryote.

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(b) Malaria is an infectious disease caused by organisms belonging to the genus *Plasmodium*.

(i) State the term used to describe an organism that causes disease.

..... [1]

(ii) Name **one** of the species of *Plasmodium* that causes malaria.

Plasmodium [1]

(c) Outline the similarities **and** differences between the modes of transmission of malaria and sleeping sickness.

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[Total: 11]